Parameter Registry — Knotted Coil Designer

Every number that shapes a result, with its true origin. Tags follow the research-integrity code standard: **LIT** (from verified literature) · **CONV** (convention/library default) · **JUDG** (our judgement) · **TUNED** (searched) · **DERIVED** (computed from others). **PLACEHOLDER** marks a JUDG value awaiting Reza's real specification.

Last updated 2026-10-20.

Physical inputs (placeholders until Reza supplies them)

Parameter	Value	Tag	Rationale / source	Where
Wire diameter d_w	1.0 mm	JUDG PLACEHOLDER	Round number for magnet-wire-scale loops. Awaiting spec (Design D5)	app, tests
Insulation thickness t_i	0.25 mm	JUDG PLACEHOLDER	Awaiting spec (D5)	app, tests
Minimum bend radius	2 mm	JUDG PLACEHOLDER	Awaiting spec (D5)	app, tests
Helmet semi-axes a, b, c	85, 105, 110 mm	JUDG PLACEHOLDER	Rough adult head plus clearance, not measured. Awaiting former dimensions (D4)	`core/surface.js`
Lowest covered height zCut	0 mm	JUDG PLACEHOLDER	Ellipsoid equator	`core/surface.js`
Face opening half-angle / top	40° / 60 mm	JUDG PLACEHOLDER	Visual guess	`core/surface.js`

Design defaults (Design.md v1.0)

Parameter	Value	Tag	Rationale / source	Where
Bloom shape	outward (lobes extend from a fixed base circle)	JUDG	Design v1.2 (Reza 2026-10-20); 'symmetric' (v1.0) selectable	knot, app
Petal sharpness m	1 (range 1–16)	JUDG	Design v1.3 (Reza 2026-10-20). Profile exponent; 1 = v1.2 shape. Range upper bound is judgement: bends tighten quickly beyond about 10	knot, app
Lobe aspect-ratio definition	length ÷ width at half length	DERIVED	Claude's definition; formula in `core/knot.js` `lobeAspect`	knot, app
Bloom range β	0.02–3.0 outward, 0.02–0.9 symmetric (UI allows 0.005–3)	JUDG	Design §3.3. **Validation run 1 showed 0.02 is below the buildable floor for the placeholder wire**. See DECISIONS.md	knot, app
Decoupling threshold |k|	0.01	JUDG PLACEHOLDER	Design D6; not yet justified from literature	`core/optimise.js`
Neighbour definition	nearest ring (d < 1.35 × pair-null spacing)	JUDG	Design v1.1 §5.1 (Reza, 2026-10-20). The v1.0 rule (footprints overlap, d < 2 Rf) is selectable and always reported	`core/optimise.js`
Bloom mode	constant core	JUDG	Design D2	app
Rotation mode	free per element	JUDG	Reza 2026-10-20: rotation as a decoupling lever	`core/optimise.js`

B1 quick look (Design v1.5)

Parameter	Value	Tag	Rationale / source	Where
Head surface	helmet semi-axes − 10 mm	JUDG PLACEHOLDER	Stand-in for a head; replace with a real head model when simulating	`core/field.js`
B1 units factor	100 µT/A per (1/mm)	DERIVED	µ0/4π with mm units; header of `core/field.js` ; checked by V8	`core/field.js`
Wire thinning for B1	≤ 200 points per element	JUDG	Speed. Field points are ≥ 10 mm from wires, so the midpoint rule is adequate (not separately tested)	`core/field.js`
Colour scale	−30 to 0 dB relative to the maximum shown, viridis	CONV	Viridis is a standard perceptually uniform colour map	app
Array combination	$\sqrt{\Sigma\lvert B1_i\rvert^2}$ (1 A per element)	CONV	Common quick stand-in for combined array sensitivity; ignores noise correlation	app

Numerical choices

Parameter	Value	Tag	Rationale / check	Where
μ0/4π	1e-7 H/m = 0.1 nH/mm	DERIVED	SI definition	`core/inductance.js`
Near-field refinement distance	3 segment lengths	JUDG	Checked indirectly by V1, V2, V4	`core/inductance.js`
Max sub-pieces per segment	12	JUDG	Same	`core/inductance.js`
Kernel softening between elements	ε = d_w + 2 t_i	JUDG	Wires crossing between elements are separated by at least this. Sensitivity not yet tested	`core/optimise.js`
Self-inductance method	centreline vs copy offset by wire radius	LIT (method)	High-frequency surface-current formula, Rosa & Grover 1912 eq. [62], VERIFIED. Checked by V2	`core/inductance.js`
Element points (element view)	adaptive: 100·p·q (min 400), doubled until crossing count is stable	JUDG	Added after validation run 1	`core/knot.js`
Stage-1 table	128 pts; d = 0.3–3.0 Rf in 55 steps; 12 rotation steps per 2π/q	JUDG / DERIVED	Speed. Stage 2 corrects table error	`core/optimise.js`
Stage-2 / final points	160 / 300 per element	JUDG	V4: k converged to < 0.001 at 300	`core/optimise.js`
Self-L points	300	JUDG		`core/optimise.js`
Worst-pair weight λ_max	1 (cost term $\lambda_{max} \cdot k^4/thr^2$)	JUDG	Chosen before testing, not tuned. Made default by the pre-registered comparison, `validation/compare_worstpair.html`	`core/optimise.js`
Coverage weight λ	1	JUDG	Makes gaps dominate k^2 terms	`core/optimise.js`
Neighbour weight taper	2.0 → 2.4 Rf (footprint); 1.35 → 1.55 × null spacing (ring1)	JUDG	Stops pairs escaping by stepping just past the cut-off	`core/optimise.js`

Parameter	Value	Tag	Rationale / check	Where
Coverage samples	2400 on full ellipsoid	JUDG		`core/optimise.js`
Search sweeps	≤ 300 (stage 1), ≤ 12 (stage 2)	JUDG	Speed vs quality; not tuned	`core/optimise.js`
Random seed	1	JUDG	Reproducibility; stored in every export	`core/optimise.js`
Bloom floor formula	c / (R sin(π/p)) outward; c / (2 R sin(π/p)) symmetric	DERIVED	Claude's geometric argument; agrees with the numerical lift solver (see DECISIONS.md)	`core/knot.js`